Question		Answer	Marks	AO	Guidance
5	(a)	$-7-1.75=(5+2)a$	M1	1.1a	Uses the combined mass in N2L equation. Allow sign errors. No extra forces. Do not allow for mga.
			A1	1.1	Correct equation Also allow $8.75=7a$ soi.
		$a=-\frac{8.75}{7}=-1.25$ ms^{-2}	A1	1.1	Cao. Negativity must follow from their working, or clearly states deceleration of 1.25 ms^{-2}
		Alternative method			
		For 5kg block $T-7=5a$			
		For 2kg block $-T-1.75=2a$	M1		Sets up an equation for each block including Tension. Allow sign errors. No extra forces. Do not allow for mga.
		$-7-1.75=(5+2)a$	A1		Correct equation with T eliminated. Any form. Also allow $8.75=7a$ soi.
		$a=-\frac{8.75}{7}=-1.25$ ms^{-2}	A1		Cao. Negativity must follow from their working, or clearly states deceleration of 1.25 ms^{-2}
			[3]		
	(b)	Model the resistance with a variable force	B1	3.5c	Allow a particular example of a variable force or variation in some other factor
			[1]		